Ogbonna Arinze

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SUMMARY

Senior Fullstack Engineer with 8 years of professional experience building high-performance, user-centric web applications across e-commerce, healthcare, fintech, and civic tech domains. Proven expertise in architecting modern frontend systems using React, Next.js, and TypeScript, with strong backend capabilities including GraphQL, NodeJS APIs, NestJS, and PostgreSQL. Track record of optimizing platform performance, rendering times, and data fetching strategies, while implementing automated testing frameworks (Playwright, Jest, Cypress) and CI/CD pipelines. Experienced in mentoring developers, conducting code reviews, and contributing to architectural decisions that drive scalable, high-quality product delivery.

CORE COMPETENCIES

- **Frontend Development:** React, Next.js, TypeScript, Vue.js, Redux, Zustand, Tailwind CSS, Radix UI, SSR/SSG, and component-based UI systems.
- **Backend & API:** GraphQL, NestJS, NodeJS REST APIs, PostgreSQL, MongoDB, Prisma, JWT authentication, and RBAC.
- **Performance & Optimization:** Code splitting, lazy loading, CDN strategies, Core Web Vitals, Lighthouse, bundle analysis, caching, and render profiling.
- **Testing & Quality:** Playwright, Jest, Cypress, Enzyme, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA.
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Docker, GitHub Actions, CI/CD pipelines, Vercel, Nx monorepo orchestration, and Webpack/Vite/Rollup.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, vector search (pgvector, Chroma), and embeddings.
- **Leadership:** Code reviews, mentorship, architecture reviews, Agile/Scrum, TDD/BDD, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)

Frontend Engineer — *May 2023 - Present*

- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources, embed the widget on external sites, and query their data in real time via a conversational interface.
- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, while maintaining and extending the existing Vue.js codebase during the migration period.
- Integrated internal and external APIs using **GraphQL** and **React Query** for efficient data fetching, implementing request caching, optimistic updates, and streamlined data flow across the application.
- Optimized platform performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing rendering times significantly as measured via **Lighthouse** and Chrome DevTools.
- Implemented **JWT-based authentication** and **role-based access control** in **Next.js** with **TypeScript**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, creating a component-based UI system that improved consistency, accessibility (WCAG), and developer velocity across product teams.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), conducting regular code reviews and fostering a culture of technical excellence.

Descasio — Lagos, Nigeria (Hybrid)

Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application, significantly boosting speed and user experience.
- Enhanced development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports.
- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices.

- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, conducting code reviews and fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume bank **Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical **E-commerce** and payment modules (MasterCard, VISA, Express-cash) using **React, Redux, RxJS**, and **Axios** for asynchronous API integration.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design** for user role and permission management.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models, focusing on testable, maintainable, and well-documented code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app (22 chart types with Recharts), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Built a **multi-agent AI system** using **Mastra** with 5 specialized agents and a three-stage intent classification pipeline, powered by a **hybrid RAG pipeline** combining **pgvector** + BM25 full-text search with Reciprocal Rank Fusion and **Cohere** reranking.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, implemented **SSE streaming** with keepalive heartbeats, and built shareable public conversations with SEO optimization (Open Graph, JSON-LD, SSR).
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, and memory limits.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*